CHAPTER 5

Trigonometric functions

S75, S76, S78, S79

Before you begin · Unit circle

Without a calculator, in six minutes. Answer briefly. Use radians, signs and differentiation. Then choose the targeted review indicated under the questions.

1. Convert 180 $^\circ$ to radians.
2. Give sin 0 and cos 0.
3. Give the sign of sin x on $]0;\pi[$.
4. Solve cos $x = 0$ on $[0;2\pi]$.
5. Give cos $(\pi/3)$.
6. Give the period of sin.

Answers

π ; 0 and 1; positive; $\frac{\pi}{2}$ & $\frac{3\pi}{2}$; $1/2$; 2π.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Review a half-turn on the circle; R85. **2.** Read the x-coordinate and y-coordinate of the point (1;0). **3.** Read the sign of the y-coordinate on the upper semicircle; R86. **4.** Locate the two points with x-coordinate zero. **5.** Use a equilateral triangle cut in half. **6.** Go round a full turn and compare two y-coordinates; R85. The R questions serve as review material with the teacher, even if their whole-class routine is scheduled later.

Objectives

Use parity and periodicity; differentiate sine and cosine; solve equations and inequalities; study a periodic function.

Course notes

Circle and functions

On the trigonometric circle, the point identified by an angle x in radians has coordinates $(\cos x, \sin x)$. For every real x,

$$\cos^2 x + \sin^2 x = 1, \quad -1 \leq \cos x, \sin x \leq 1.$$

Cosine is even: $\cos(-x) = \cos x$. Sine is odd: $\sin(-x) = -\sin x$. The two functions are 2π-periodic.

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

Derivatives and variations

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

$$(\sin(ax+b))' = a\cos(ax+b), \quad (\cos(ax+b))' = -a\sin(ax+b).$$

On $[-\pi, \pi]$, cosine increases on $[-\pi, 0]$ from -1 to 1, then decreases on $[0, \pi]$ from 1 to -1. Sine decreases on $[-\pi, -\frac{\pi}{2}]$, increases on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, then decreases on $[\frac{\pi}{2}, \pi]$.

Justification. Read the sign of sine and cosine on the circle, then apply the link between derivative and variations. Periodicity extends the curves; it does not remove the need to study an interval of sufficient length.

Equations and inequalities

If $|a| > 1$, $\cos x = a$ and $\sin x = a$ have no real solution. For a known angle θ,

$$\cos x = \cos\theta \iff x = \theta + 2k\pi \text{ or } x = -\theta + 2k\pi,$$

$$\sin x = \sin\theta \iff x = \theta + 2k\pi \text{ or } x = \pi - \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

The solutions are then restricted to the required interval and duplicates are removed.

Examples on $[-\pi, \pi]$. $\cos x = \frac{1}{2}$ gives $x = \pm\frac{\pi}{3}$; $\sin x = \frac{1}{2}$ gives $x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. $\cos x \leq \frac{1}{2}$ gives $[-\pi, -\frac{\pi}{3}] \cup [\frac{\pi}{3}, \pi]$; $\sin x \geq \frac{1}{2}$ gives $[\frac{\pi}{6}, \frac{5\pi}{6}]$.

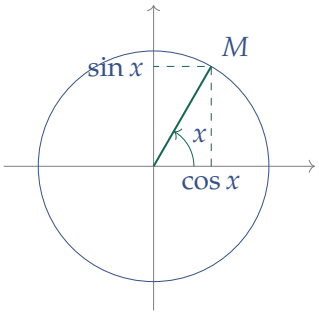
A trigonometric composite

To solve $\cos(2x) = \frac{1}{2}$ on $[-\pi, \pi]$, set $t = 2x \in [-2\pi, 2\pi]$, solve for t, then divide by 2. We get $x \in \{-\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}\}$.

Optimisation. For $f(x) = \sin x + \frac{1}{2}\sin(2x)$ on $[0, \pi]$, $f' = \cos x + \cos(2x) = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1)$. The function increases up to $\frac{\pi}{3}$ then decreases; maximum $\frac{3\sqrt{3}}{4}$.

Double-angle identities used in the exercises. For every real x, $\sin(2x) = 2\sin x\cos x$ and $\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1$. These identities are recalled here so that their use in the problems is not left implicit.

Curve symmetries. The parity of cosine corresponds to a symmetry with respect to the y-axis; the oddness of sine to a point symmetry about the origin. Periodicity extends each curve by translation by the vector $(2\pi, 0)$.



The coordinates of the point on the circle are $(\cos x, \sin x)$.

From the circle to the study of a function

5.1 One directed angle, two coordinates

First use: S75.

Definition · Sine and cosine

On the unit circle oriented anticlockwise, the point at angle x has coordinates $(\cos x; \sin x)$. Adding 2 $k\pi$ with $k \in \mathbb{Z}$ gives the same point. A negative measure corresponds to the opposite direction; it is not a negative length.

Worked example · Reading an equation and an inequality

On $[-\pi;\pi]$, $\cos x = 1/2$ corresponds to the two points with x-coordinate 1/2: $x = \pm\pi/3$. The inequality $\cos x \leq 1/2$ selects the arc to the left of the vertical line x = 1/2: $[-\pi;-\pi/3] \cup [\pi/3;\pi]$. The endpoints belong to the solution because the inequality is non-strict.

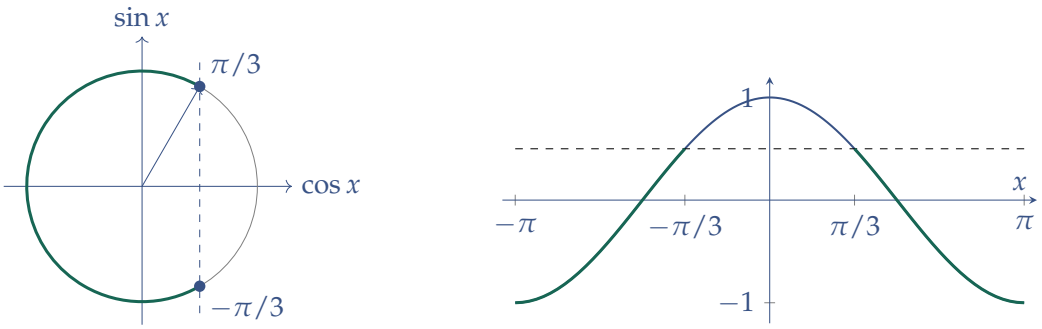


Figure 5.1. Same condition $\cos x \leq 1/2$: arc to the left of the vertical on the circle, curve below level 1 /2 to right.

A1 · Immediate application Solve $\sin x = 1/2$ on $[0;2\pi[$, then $\sin x \geq 1/2$ on that same interval. Show the solutions on a circle and on a curve.

5.2 Reading amplitude and period

First use: S76.

Worked example · A shifted oscillation

For $h(t) = 3 + 2\sin(4t)$, the values lie between 1 and 5: the mean level is 3, the amplitude is 2. The smallest positive period is $2\pi/4 = \pi/2$. The number 2π remains a period, but is not the smallest.

A2 · Partial guidance For $g(t) = 1 - 3\cos(2t)$, give the minimum, maximum, amplitude and smallest positive period. Find the values of t in $[0;\pi]$ where the extrema are attained.

A3 · Refute "The smallest positive period of $\sin^2 x$ is 2π, like that of $\sin x$." Show that a translation by π leaves the function unchanged; explain why $\pi/2$ is not a period.

5.3 The sign comes from the circle

First use: S78.

Property · Derivatives and antiderivatives

The derivatives are $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$. For a composite, multiply by the inner derivative. An antiderivative of $\cos(ax+b)$ for $a \neq 0$ is $\sin(ax+b)/a$; an antiderivative of $\sin(ax+b)$ is $-\cos(ax+b)/a$.

Worked example · Variation on an interval

For $f(x) = x + \cos x$ on $[0;\pi]$, $f' = 1 - \sin x \geq 0$, with equality only at $\pi/2$. The function is strictly increasing: it strictly increases on each side of this isolated point and remains continuous. Moreover $f'' = -\cos x$ changes sign at $\pi/2$, so the curve has an inflection at $(\pi/2;\pi/2)$. A horizontal tangent is not necessarily an extremum.

A4 · Choice of strategy Study $f(x) = x - \sin x$ on $[0;\pi]$: derivative, variations, convexity. Deduce $\sin x \leq x$ on this interval. Does the formula $f'(0) = 0$ imply a change in variation?

5.4 An identity put to work on an area

First use: S79.

Worked example · Area of a rectangle

A point in the first quadrant of the unit circle has coordinates $(\cos\theta;\sin\theta)$, $0 \leq \theta \leq \pi/2$. The rectangle symmetric about the axes has area $A(\theta) = 4\sin\theta\cos\theta$. Since $(\sin\theta - \cos\theta)^2 \geq 0$ and $\sin^2\theta + \cos^2\theta = 1$, we have $2\sin\theta\cos\theta \leq 1$, therefore $A \leq 2$. Equality requires $\theta = \pi/4$: the optimal rectangle is a square.

A5 · Synthesis Determine by differentiation the maximum of $4\sin\theta\cos\theta$ on $[0;\pi/2]$. Recover the previous conclusion, then calculate $\int_0^{\pi/2} \sin\theta\cos\theta\,\mathrm{d}\theta$.

A6 · Independent problem Prove that $\cos x = x$ has a unique solution on $[0;\pi/2]$. Bound it between 0.7 and 0.8 using the calculator in radian mode. Suggest a method to improve the precision.

A first step: Study the difference of the two sides and separate existence, uniqueness and approximation.

One minute, spoken aloud. Explain how the same set of solutions appears as arcs on the circle and intervals on the x-axis of the curve.

Delayed review · S84 then S90. Mix limits, differentiation, equations, areas and trigonometry in a single study; redo a solution on another interval, checking the endpoints.

Investigation

Height of a point on a wheel - 35 min.

A wheel turns at constant speed in a vertical plane. Its radius is 2 m and its centre is 3 m above the ground. A point on the wheel is at its lowest position at time $t = 0$. The wheel completes one revolution in 8 s. Its height is modelled by $h(t) = 3 - 2\cos\left(\frac{\pi t}{4}\right)$, where t is expressed in seconds and $h(t)$ in metres.

1. Check that the model gives the stated initial height, a period of 8 s and the heights expected minimum and maximum.
2. Calculate $h'(t)$, then study the variations of h on $[0;8]$.
3. Determine all the times $t \in [0;8]$ for which the height is 4 m.
4. Determine the total time, during this revolution, for which the point is at least 4 m off ground.

Progressive exercises

E1 · Direct application. Solve in $[-\pi;\pi]$ the equation $\sin x = -\frac{1}{2}$.

E2 · Application and justification. Solve in $[-\pi;\pi]$ the inequality $\cos x \geq \frac{\sqrt{2}}{2}$.

E3 · Method to organise. We define on $\mathbb{R}$ the functions $f(x) = \cos(3x-1)$ and $g(x) = \sin^2 x$. Calculate their derivatives.

E4 · Choice of strategy. Consider $f(x) = 2 + \sin x$ on $[0;2\pi]$. Calculate its derivative and draw its table of variations.

E5 · Synthesis or proof. Calculate $\int_0^{\pi} \sin x \, \mathrm{d}x$, then the average value of the sine function on $[0;\pi]$.

Work from the textbook or a past paper

Barbazo, p. 166, Exercise 19. In S75: question 1, to read a symmetry and a period on a curve. At home: question 2, to prove the conjectures using identities valid for all reals.

Homework

DM5 - Splitting an area - 55 min, after S79.

Consider the function f defined on $[0;\frac{\pi}{2}]$ by $f(x) = \sin x \cos x$. Its graph is drawn in a orthonormal coordinate system.

1. Calculate $f'(x)$ and determine the maximum of f over this interval.

2. Solve on $[0;\frac{\pi}{2}]$ the inequality $f(x) \geq \frac{1}{4}$.

3. Calculate $\int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x$, then the average value of f on $[0;\frac{\pi}{2}]$.

4. For $a \in [0;\frac{\pi}{2}]$, set $A(a) = \int_0^a f(x)\,\mathrm{d}x$. Determine a so that the region under the curve between 0 and a has an area equal to half the total area under the curve between 0 and $\frac{\pi}{2}$.

MCQ A · circle and functions

After S76; k denotes an integer.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $\cos(-x)$ is:

A. $-\cos x$ B. $\cos x$

C. $\sin x$ D. 0

2. The smallest strictly positive period of $x \mapsto \sin(3x)$ is:

A. 2π B. 3π

C. $\frac{2\pi}{3}$ D. $\frac{\pi}{3}$

3. The solutions of $\sin x = 0$ are:

A. $x = \frac{\pi}{2} + 2k\pi$ B. $x = k\pi$

C. $x = 2k\pi$ only D. $x = \pi/4$

4. The derivative of $\cos x$ is:

A. $\sin x$ B. $-\sin x$

C. $-\cos x$ D. $\tan x$

5. On $[0;\pi]$, the sine is:

A. always negative B. positive or zero

C. always increasing D. always zero

MCQ B · equations and function study

After S79.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. Solve $\cos x = 1$ on $[0;2\pi]$ gives:

A. 0 and 2π B. π

C. $\frac{\pi}{2}$ D. a single solution

7. To solve $\sin x \geq \frac{1}{2}$, in particular we need:

A. choose the correct arc, then intersect with the interval required B. divide by $\sin x$

C. use a binomial D. forget the endpoints

8. A function of the form $a + b\cos(\omega t)$, with $\omega \neq 0$ has amplitude:

A. a B. b

C. $|b|$ D. ω

9. If $f'(x) = \cos x$, f is decreasing on:

A. $]0;\frac{\pi}{2}[$ B. $]\frac{\pi}{2};\frac{3\pi}{2}[$

C. $]\frac{3\pi}{2};2\pi[$ D. $[-\frac{\pi}{2};\frac{\pi}{2}]$

10. Two solutions differing by 2π represent:

A. distinct angles but the same point on the circle B. A contradiction

C. opposite sines D. forbidden values

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $\sin^2 x + \cos^2 x = 1$ for every real x.
2. $\sin x = \frac{1}{2}$ has only one solution on $[0;2\pi]$.
3. $\sin(a+b) = \sin a + \sin b$ for all real numbers a, b.

Review of prior work

I can read the circle, exploit parity and period, solve with an interval and study a function. **À**

Watch for: forgetting an endpoint solution; confusing a reading from a graph with a proof of an identity.

MCQ solutions

1. **B.** Cosine is even: $\cos(-x) = \cos x$. It is sine that is odd.
2. **C.** The smallest positive shift of the argument equal to 2π is obtained for $T = 2\pi$, i.e. $T = 2\pi/3$. 2π is also a period but is not the smallest.
3. **B.** The zeros of sine are the integer multiples of π, with $k \in \mathbb{Z}$. Keeping only $k\pi$ loses the odd multiples.
4. **B.** The derivative of cosine is minus sine, with angles in radians.
5. **B.** Sine is positive or zero on the upper semicircle. It increases up to $\pi/2$ then decreases; it is not increasing on the whole interval.
6. **A.** The two endpoints 0 and 2π are two distinct numbers in the interval, even if they correspond to the same point on the circle.
7. **A.** Identify the arc where the y-coordinate is at least $1/2$, then apply periodicity and the bounds of the interval. Dividing by a sine whose sign varies would be unjustified.
8. **C.** For $\omega \neq 0$, the extreme values are $a - |b|$ and $a + |b|$. The amplitude is half their difference, i.e. $|b|$.
9. **B.** Cosine is negative on $]\pi/2;3\pi/2[$; the sign of the derivative gives the decrease of f.
10. **A.** Adding a complete turn keeps the same point on the circle, as well as sine and cosine. The angles remain two distinct numbers.

True/false. True: the point on the unit circle has coordinates $(\cos x ; \sin x)$ and the sum of squares is 1. False: there are $\pi/6$ and $5\pi/6$. False: for $a = b = \pi/2$, the left-hand side equals 0 and the right-hand side 2.

Solutions to the applications and guidance

Application 1. Equation: $\pi/6$ and $5\pi/6$. Inequality: $[\pi/6; 5\pi/6]$, upper arc above the level y = 1/2.

Application 2. Minimum -2 at t=0 and π ; maximum 4 at $\pi/2$; amplitude 3; smallest positive period π. The minus sign swaps the positions of the maxima and minima of cosine.

Application 3. $\sin(x+\pi) = -\sin x$, so the squares are equal. But the values at 0 and $\pi/2$ are 0 and 1: $\pi/2$ is not a period. To prove the minimality of π, a period T>0 must satisfy $\sin^2 T = 0$, so T is a positive integer multiple of π.

Application 4. $f' = 1 - \cos x \geq 0$, strictly positive on $]0;\pi]$, so f is increasing, strictly after 0. $f'' = \sin x \geq 0$, hence convexity. Since f(0)=0, $x - \sin x \geq 0$. A zero of the derivative at an endpoint does not force any change in variation.

Application 5. $A' = 4(\cos^2\theta - \sin^2\theta)$, positive before $\pi/4$ and negative after, hence maximum 2. A antiderivative of $\sin$ $\theta\cos\theta$ is $\frac{1}{2}\sin^2\theta$, so the integral equals 1/2.

Application 6. $G(x) = \cos x - x$ is continuous, $G(0) = 1 > 0$, $G(\pi/2) = -\pi/2 < 0$; the IVT gives one solution. $G' = -\sin x - 1 < 0$ on the interval ensures uniqueness. $G(0{,}7) \simeq 0{,}0648 > 0$, $G(0{,}8) \simeq -0{,}1033 < 0$. Bisection on this interval improves the bounds.

Teaching notes

Have students set radian mode explicitly. For optimisation, compare the algebraic proof and differentiation: the identity explains the bound, the derivative describes the variations. Revisit arcs with a half-open interval to handle duplicates.

Teaching notes and solutions

Guidance for the teacher

This chapter actually ends in early May, not late April. It is therefore not included in the second April mock exam. Revise the circle before the curves. Use radians in the derivatives. Have students check the bounds $-\pi$ and π : they represent the same point of the circle but are two distinct numbers if the interval contains them. The variations of composite functions are deduced from the full sign of the derivative.

Solutions to the activity

1. $h(0) = 3 - 2 = 1$ m. Adding 8 to t adds 2 π to the argument of the cosine: 8 s is a period, and it is the smallest positive period. Since $-1 \leq \cos \leq 1$, $1 \leq h \leq 5$; both bounds are attained.

2. By differentiating a composite, $h'(t) = (\frac{\pi}{2})\sin(\frac{\pi t}{4})$. The argument ranges over $[0, 2\pi]$. The sine is positive on $]0, \pi[$, then negative on $]\pi, 2\pi[$. Thus h increases from 1 to 5 m over $[0, 4]$, then decreases from 5 to 1 m on $[4, 8]$.

3. $h(t) = 4 \iff \cos(\frac{\pi t}{4}) = -\frac{1}{2}$. On $[0, 2\pi]$, the corresponding angles are $\frac{2\pi}{3}$ & $\frac{4\pi}{3}$; therefore $t = \frac{8}{3}$ s or $t = \frac{16}{3}$ s.

4. The condition $h(t) \geq 4$ is equivalent to $\cos(\frac{\pi t}{4}) \leq -\frac{1}{2}$, hence to $t \in [\frac{8}{3}, \frac{16}{3}]$. The duration is $\frac{16}{3} - \frac{8}{3} = \frac{8}{3}$ s.

E1. On the circle, the sine is $-\frac{1}{2}$ at angles $-\frac{\pi}{6}$ & $-\frac{5\pi}{6}$ in $[-\pi, \pi]$. The other angles are obtained by adding 2 $k\pi$ and lie outside the required interval.

E2. The cosine is at least $\frac{\sqrt{2}}{2}$ on the arc around the angle 0, between $-\frac{\pi}{4}$ & $\frac{\pi}{4}$, endpoints included. The solution set is therefore $[-\frac{\pi}{4}, \frac{\pi}{4}]$.

E3. By composition, $(\cos(3x - 1))' = -3\sin(3x - 1)$. For the square of sine, $(\sin^2 x)' = 2\sin x \cos x$. The two derivatives are defined on $\mathbb{R}$.

E4. $f'(x) = \cos x$, positive on $[0, \frac{\pi}{2}[$ and $]\frac{3\pi}{2}, 2\pi]$, negative on $]\frac{\pi}{2}, \frac{3\pi}{2}[$. Thus f increases from 2 to 3, decreases from 3 to 1, then increases from 1 to 2. The maximum is 3 at $\frac{\pi}{2}$ and the minimum 1 at $\frac{3\pi}{2}$.

E5. An antiderivative of sine is $-\cos x$. Hence the integral equals $[-\cos x]_0^{\pi} = 1 - (-1) = 2$. The mean value over this interval of length π is $\frac{2}{\pi}$.

Solutions to homework DM5

1. By the product rule, $f' = \cos^2 x - \sin^2 x = \cos 2x$. On the interval studied, 2 $x \in [0, \pi]$: the derivative is positive before $x = \frac{\pi}{4}$ and negative afterwards. The maximum is $f(\frac{\pi}{4}) = (\frac{\sqrt{2}}{2})^2 = \frac{1}{2}$.

2. Thanks to $\sin 2x = 2\sin x \cos x$, $f(x) \geq \frac{1}{4}$ is equivalent to $\sin 2x \geq \frac{1}{2}$. For 2 $x \in [0, \pi]$, this gives $2x \in [\frac{\pi}{6}, \frac{5\pi}{6}]$, therefore $x \in [\frac{\pi}{12}, \frac{5\pi}{12}]$.

3. The derivative of $\frac{1}{2}\sin^2 x$ is $\sin x \cos x = f(x)$. Hence $\int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x = [\frac{1}{2}\sin^2 x]_0^{\frac{\pi}{2}} = \frac{1}{2}$. The mean value is $\frac{\frac{1}{2}}{\frac{\pi}{2}} = \frac{1}{\pi}$.

4. $A(a) = \frac{1}{2}\sin^2 a$. Half of the total area is $\frac{1}{4}$, therefore $\sin^2 a = \frac{1}{2}$. On $[0, \frac{\pi}{2}]$, sine is positive and strictly increasing: this equality is equivalent to sin $a = \frac{\sqrt{2}}{2}$, then to $a = \frac{\pi}{4}$.

Solutions to the textbook or past-paper work

The curve of $f(x) = \cos(3x)$ suggests a symmetry about the y-axis and a repetition after a horizontal translation of $\frac{2\pi}{3}$. For every $x \in \mathbb{R}$, $f(-x) = \cos(-3x) = \cos(3x) = f(x)$: f is even. On the other hand,

$$f\left(x + \frac{2\pi}{3}\right) = \cos(3x + 2\pi) = \cos(3x) = f(x).$$

The number $\frac{2\pi}{3} > 0$ is therefore a period of f. The drawing suggests these properties; the identities prove them for every real.